

COMP 6915

Introduction to Machine Learning Winter 2026



Department of Computer Science

Instructor: Dr. Karteek Popuri
Office: EN-2014
Office Hours: Tue, Thu @ 5:00 p.m. – 6:00 p.m. (by appointment)
e-mail: kpopuri@mun.ca

Course Description:

This course introduces concepts and algorithms in machine learning for regression and classification tasks. The course gives the student the basic ideas and intuition behind model selection and evaluation, and selected machine learning methods such as random forests, support vector machines, and neural networks.

Pre-requisites:

- Python programming, a statistics course at the undergraduate level (similar to STAT 2550), and a linear algebra course at the undergraduate level (similar to MATH 2050).

Course Objectives:

- Understand the basis underlying supervised machine learning methods.
- Understand the algorithms behind certain machine-learning methods.
- Acquire hands-on experience applying supervised machine-learning methods.
- Be able to appropriately evaluate and select a machine-learning model for a specific task.

Textbooks:

- [*The Elements of Statistical Learning: Data Mining, Inference, and Prediction*](#) by Hastie T., Tibshirani R., Friedman J.H., Springer, 2009. [ESL]
- [*Probabilistic Graphical Models: Principles and Techniques*](#) by Koller, D., Friedman, N., 2009. [PGM]
- [*Deep Learning*](#) by Goodfellow I., Bengio Y., Courville A., MIT Press, 2016. [DL]
- [*Mathematics for Machine Learning*](#) by Deisenroth M. P., Faisal A. A., Ong C. S., Cambridge University Press, 2020. [MML]
- [*An introduction to statistical learning: with Applications in Python*](#) by James G., Witten D., Hastie T., Tibshirani R., Taylor J., Springer, 2023. [ISL]
- [*Understanding Deep Learning*](#) by Prince, S. J. D., The MIT Press, 2023. [UDL]

Evaluation:

The final grade in this course will be determined as follows:

Assignments (5)	40%
Quizzes (10)	10%
Literature review	20%
Project	30%

Format:

Lectures, three hours per week.

Lecture Time: Tue, Thu 3:30 p.m. - 4:45 p.m.

Lecture Room: EN-2043

PLEASE NOTE: Course information (e.g., assignments, lecture notes, announcements, grades, etc.) can be found on Brightspace (D2L). However, please note that your instructional team **DOES NOT** check email in Brightspace (D2L). Any e-mail messages to the instructional team should begin with the tag **[COMP6915-W26]** in the subject line and must be sent from a valid Memorial University email account. Please note that you **cannot** email from a D2L “online.mun.ca” email address to a “mun.ca” email address. Thus, D2L “online.mun.ca” email addresses can only be used within the D2L environment.

Course Schedule (Tentative):

Dates	Lecture #	Chapter(s)	Topics(s)
Week 1: Jan 5 – Jan 9	1, 2		Introduction
Week 2: Jan 12 – Jan 16	3, 4	ISL-2; ESL-2; DL-5; MML-8	Machine Learning basics
Week 3: Jan 19 – Jan 23	5, 6	ISL-2; ESL-2	k-Nearest Neighbors (kNN)
Week 4: Jan 26 – Jan 30	7, 8	ISL-5; ESL-7	Model Assessment and Selection
Week 5: Feb 2 – Feb 6	9, 10	ISL-3, 6; ESL-3 MML-9	Linear Methods for Regression
Week 6: Feb 9 – Feb 13	11, 12	ISL-4; ESL-4	Linear Methods for Classification
Week 7: Feb 16 – Feb 20	13, 14	ISL-9; ESL-12; MML-12	Support Vector Machines (SVMs)
Week 8: Feb 23 – Feb 27	Winter semester break		
Week 9: Mar 2 – Mar 6	15, 16	ISL-8; ESL-9, 15	Tree-based methods
Week 10: Mar 9 – Mar 13	17, 18	PGM-3	Probabilistic Graphical models
Week 11: Mar 16 – Mar 20	19, 20	UDL-4; DL-6; ISL-10; ESL-11	Feedforward Networks
Week 12: Mar 23 – Mar 27	21, 22	UDL-10; ISL-10	Convolutional Neural Networks (CNNs)
Week 13: Mar 30 – Apr 3	23, 24	UDL-11	Residual Neural Networks
Week 14: Apr 6 – Apr 10	25	MML-10; ISL-12; ESL-14	Unsupervised Machine Learning

In-class Quizzes Schedule (Tentative):

Quiz #	Date
1	Thursday, Jan 22
2	Thursday, Jan 29
3	Thursday, Feb 5
4	Thursday, Feb 12
5	Thursday, Feb 19
6	Thursday, Mar 5
7	Thursday, Mar 12
8	Thursday, Mar 19
9	Thursday, Mar 26
10	Thursday, Apr 2

Assignment Schedule (Tentative):

Assignment #	Assignment Due at 11:59 p.m. NT
1	Monday, Feb 2
2	Monday, Feb 16
3	Monday, Mar 2
4	Monday, Mar 16
5	Monday, Mar 30

Literature review and Project submission Schedule (Tentative):

Submission	Due at 11:59 p.m. NT
Literature review – Topic	Tuesday, Feb 3
Project – Proposal	Tuesday, Feb 3
Literature review – Report	Tuesday, Apr 21
Project - Report	Tuesday, Apr 21

Important notes to read, understand and follow for this course:

• General Notes

- Memorial University of Newfoundland is committed to fostering equitable and accessible learning environments for all students. Accommodations for students with disabilities are provided in accordance with the [Accommodations for Students with Disabilities Policy](#) and its related procedures. Students who feel that they may require formal academic accommodations to address barriers or challenges they are experiencing related to their learning are encouraged to contact [Accessibility Services \(the Blundon Centre\)](#) at the earliest opportunity to ensure any required accommodations are provided in a timely manner. You can contact Accessibility Services (the Blundon Centre) by emailing blundon@mun.ca. **Students using Blundon services for exams are strongly recommended to book the exams with the Blundon Centre at least 3 weeks ahead of the exam date to allow for arrangements to be made. Students who are registered with the Blundon Centre are also encouraged to contact the course instructor, during a lab period, to discuss accommodations.**
- Academic integrity means taking full responsibility for the academic work you submit for your courses, so that your professors can evaluate you based on your own understanding and effort. It means being honest and honorable in all academic pursuits, even in difficult circumstances. Students are expected to know and avoid academic offences; ignorance of an offence is not an acceptable excuse for committing it. Penalties could include reprimand, reduction of grade, probation, suspension, or expulsion from the University. For more information, refer to the University Regulations for Academic Misconduct (Section 6.12) in the University Calendar, revisit the INTG 1000 course in Brightspace, or see the [undergraduate page about academic integrity](#).
- Course information (e.g., assignments, quizzes, links to lecture notes, announcements, grades, etc.) can be found in Brightspace (D2L).
- All quizzes are to be done **individually**.
- **Assignments, Literature review and Project work are to be done by individuals from a single team.**
- A doctor's note is not required for short term illness. Please refer to the **current University policy** regarding medical notes and the information to be in them. For more information, please see the University Calendar - University Regulations - General Academic Regulations (Undergraduate) - **6.7.5 Exemptions from Parts of the Evaluation** and **6.15 Appeal of Decisions** or consult the Registrar's Office.
- **The lectures and displays (and all material) delivered or provided in COMP 6915 – Introduction to Machine Learning by Karteeek Popuri, including any visual or audio recording thereof, are subject to copyright.** It is prohibited to record or copy by any means, in any format, openly or surreptitiously, in whole or in part,

in the absence of express written permission from Karteek Popuri any of the lectures, materials provided or published in any form during or from the course.

- Please note that the last day to drop a course without academic prejudice, this semester, is **March 3rd**. You do not need anyone's permission to drop the course before the last drop date. You can drop the course yourself using Memorial Self-Service. If you have any difficulties with dropping the course at that time, then please contact reghelp@mun.ca. Anyone wanting to drop the course after that deadline should contact our undergraduate advisor, at cs-ugradadv@mun.ca, to make the request.
- **IMPORTANT:** Late drops can only be approved in very specific circumstances, as detailed in Regulation 6.5.6.2 here: <https://www.mun.ca/regoff/calendar/sectionNo=REGS-0524>.

- **COVID Related Notes**

- In the case of a class disruption or cancellation, and in the case of revisions to evaluation methods, the instructor or the Head of the department will notify all students registered in the course via the course shell in Brightspace. Any necessary revisions to the evaluation methods will be made in consultation with the students registered in this course. If a student demonstrates that they would be disadvantaged by the change, then, as per 6.7.4 of the University Calendar, accommodations will be made.
- Masks are currently not required on campus. If this policy changes during the term, you may be required to use a mask. There will be different levels of comfort and anxiety for mask use. Should other health directives or the overall situation connected to COVID-19 change over the course of the term, a back-up plan for remote delivery is in place to ensure that the course will continue and to minimize disruption to the student experience.
- As a part of our shared responsibility to keep each other and our extended families safe during the upcoming school year, COVID-19 vaccines are required for all students, faculty, and staff; masks are required on all Memorial campuses in all indoor spaces. For further information, see the University webpage on [COVID-19 Vaccination Clinics](#).
- There is nothing more important than your mental and physical health. Doctors' notes are not required for medical absences in this course. You are encouraged to seek appropriate medical attention from the Student Wellness and Counselling Centre. I encourage you to reach out to the Blundon Centre as early as possible to discuss any adjustments you think may be necessary for you to successfully complete this course.
- To protect yourself and those around you, it is important to stay home if you feel unwell, or if you are under quarantine because you have potentially been exposed to the virus. Please keep your instructor informed so we can work together to allow you to keep up with the course materials should you need to miss labs. You will not be penalized if you need to stay home for quarantine. Memorial University has recognized the importance of academic leniency as we work to keep our campus safe for all.
- Memorial encourages faculty, staff, and students to [download the COVID Alert app](#) to help protect yourself and others. The app is designed to let Canadians know whether they may have been exposed to COVID-19.
- While the COVID-19 pandemic is slowly subsiding in many parts of the world and vaccination rates are increasing, this is still a stressful time for many. It's important that we support each other and keep informed of current information. The [Memorial COVID-19 website](#) is an excellent source of information and support, with specific links for students, supports and services, and health and wellness.

- **Assignments**

- Assignments will be due at **11:59 p.m.** on the specified due date. **No late assignments will be accepted.** See table above for assignment due dates. **You will receive a 0 for an assignment submitted late.**
- Each assignment should be uploaded in the correct format, and in the correct folder, using Brightspace's **Dropbox**. Be aware that the files you submit for evaluation should be uploaded on or before the due date and much before the cut off time, **11:59 p.m. Newfoundland Time**. Even if you are late by a few seconds, you will not be allowed to submit your work; hence you should try to upload the files 15

minutes before the cut off time since your system clock is not synchronized with CITL's system clock and the cutoff time is based on CITL's system clock. Please note that if your file has been correctly uploaded, you will get a confirmation receipt from the **Dropbox** tool. If you do not receive this receipt, please contact the CITL Support team (<https://www.citl.mun.ca/support>). Your instructor will not be able to help you in this regard. It is **very important** to remember that what is not uploaded, cannot be marked. It is your responsibility to make sure that the **correct** files are uploaded, so please **do** check for the confirmation that your files have been uploaded.

- **Assignments are to be done by members of a single team. If cheating/plagiarism is detected for an assignment, the assignment will be given 0%. The department head will also be notified of any academic misconduct.**

- **Literature review and Project**

- Literature review and Project deliverable submissions will be due at **11:59 p.m.** on the specified due date. **No late submissions will be accepted.** See table above for assignment due dates. **You will receive a 0 for a late submission.**
- Literature review and Project deliverables should be uploaded in the correct format, and in the correct folder, using Brightspace's **Dropbox**. Be aware that the files you submit for evaluation should be uploaded on or before the due date and much before the cut off time, **11:59 p.m. Newfoundland Time**. Even if you are late by a few seconds, you will not be allowed to submit your work; hence you should try to upload the files 15 minutes before the cut off time since your system clock is not synchronized with CITL's system clock and the cutoff time is based on CITL's system clock. Please note that if your file has been correctly uploaded, you will get a confirmation receipt from the **Dropbox** tool. If you do not receive this receipt, please contact the CITL Support team (<https://www.citl.mun.ca/support>). Your instructor will not be able to help you in this regard. It is **very important** to remember that what is not uploaded, cannot be marked. It is your responsibility to make sure that the **correct** files are uploaded, so please **do** check for the confirmation that your files have been uploaded.
- **Literature review and Project are to be done by members of a single team. If cheating/plagiarism is detected for an assignment, the assignment will be given 0%. The department head will also be notified of any academic misconduct.**

- **Student Support**

- Memorial University of Newfoundland is committed to fostering equitable and accessible learning environments for all students. Accommodations for students with disabilities are provided in accordance with the [Accommodations for Students with Disabilities Policy](#) and its related procedures. Students who feel that they may require formal academic accommodations to address barriers or challenges they are experiencing related to their learning are encouraged to contact [Accessibility Services \(the Blundon Centre\)](#) at the earliest opportunity to ensure any required accommodations are provided in a timely manner. You can contact Accessibility Services (the Blundon Centre) by emailing blundon@mun.ca. **Students using Blundon services for exams are strongly recommended to book the exams with the Blundon Centre at least 3 weeks ahead of the exam date to allow for arrangements to be made. Students who are registered with the Blundon Centre are also encouraged to contact the course instructor, during a lab period, to discuss accommodations.**
- In addition to your instructor, instructional staff are also available to help students with course material through the discussion forum, via email/phone, or by appointment in an Online Room. See the **Instructional Staff Contact Information and Schedule** link on the course web page.
- The on-campus Computer Science Student Help Centre is in EN-2026, S. J. Carew Engineering Building, St. John's campus. Please see the following link (also available on the course Brightspace page) for hours of operation each semester:
https://www.mun.ca/computerscience/ugrad/HelpCtr_Schedule.php

- Help with a variety of Brightspace technical issues is available by contacting the **CITL Support Centre** via phone (1-866-435-1396, or locally at 709-864-8700), online chat (when available) or via their ticketing system. This contact information is available at: <https://www.citl.mun.ca/support/>
- Note that, while the due times for submissions are at **11:59 p.m. Newfoundland Time**, help may not be available after 4:30 p.m. on the due date. The last few hours from 4:30 p.m. – 11:59 p.m. should be used to finalize your work and submit it. Any questions you may have about the lab exercises, quizzes or assignments should be asked well in advance of the due date/time to allow time for help to be given.
- Please note that Memorial University offers a broad range of supports. For a more comprehensive list of student supports and resources, please check out the following links:
 - <https://www.mun.ca/munup/support/>
 - <https://www.mun.ca/main/students.php>
 - <https://www.munsu35.ca/resource-centres/>